

IKA, Russian Federation

WMO: 300280

Lat: 59.311N

Lon: 106.347E

Elev: 352

StdP: 97.17

Time Zone: 8.00 (E08)

Period: 82-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -47.7	(c) -44.9	(d) -49.9	(e) 0.0	(f) -47.7	(g) -47.3	(h) 0.0	(i) -44.8	(j) 7.3	(k) -14.7	(l) 6.6	(m) -15.5	(n) 0.3	(o) 180	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.8	28.6	17.9	26.6	17.1	24.6	16.2	19.0	25.7	18.0	24.9	17.0	23.0	2.4	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.7	12.4	21.3	15.6	11.6	20.2	14.5	10.7	19.4	55.7	25.6	52.4	24.9	49.4	23.2	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.2	(b) 6.3	(c) 5.4	(e) -50.7	(f) 31.7	(g) 1.7	(h) 1.4	(i) -52.0	(j) 32.7	(k) -53.0	(l) 33.5	(m) -54.0	(n) 34.3	(o) -55.2	(p) 35.3		
(a) 7.2	(b) 6.3	(c) 5.4	(e) -50.7	(f) 21.1	(g) 1.7	(h) 1.6	(i) -52.0	(j) 22.2	(k) -53.0	(l) 23.1	(m) -54.0	(n) 24.0	(o) -55.2	(p) 25.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-5.3	-27.9	-23.7	-13.6	-3.5	4.9	14.0	15.8	12.8	5.3	-5.1	-18.5	-25.3		
	DBStd	16.95	9.69	9.17	7.70	6.09	4.27	4.20	3.16	3.28	4.28	6.43	10.03	10.18		
	HDD10.0	6017	1174	944	733	406	166	13	1	12	152	469	854	1093		
	HDD18.3	8649	1433	1178	991	656	416	139	89	172	392	728	1104	1352		
	CDD10.0	432	0	0	0	0	9	132	181	99	10	0	0	0		
	CDD18.3	21	0	0	0	0	0	8	12	1	0	0	0	0		
	CDH23.3	668	0	0	0	0	7	302	283	73	3	0	0	0		
	CDH26.7	159	0	0	0	0	0	82	64	13	0	0	0	0		
(14) Wind	WSAvg	1.9	1.4	1.6	2.1	2.6	2.6	2.2	1.9	1.7	1.8	2.0	1.8	1.6		
(15) Precipitation	PrecAvg	288	9	7	8	16	27	45	58	46	27	22	15	11		
	PrecMax	394	23	15	25	36	64	83	153	92	63	43	26	23		
	PrecMin	187	2	1	2	4	8	10	9	10	11	5	7	3		
	PrecStd	59	5	4	5	8	13	21	40	21	12	9	6	5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-3.8	-1.4	7.5	14.2	24.5	31.2	30.6	28.9	23.1	10.4	4.7	-1.7		
		MCWB	-4.9	-3.2	2.6	6.4	12.6	18.6	18.5	17.9	14.7	5.1	3.1	-3.1		
	2%	DB	-7.4	-3.8	4.0	11.5	20.2	29.1	28.4	25.3	18.9	7.3	0.8	-4.4		
		MCWB	-8.3	-5.2	0.6	5.4	11.0	17.6	18.2	16.5	12.0	3.9	-0.4	-5.3		
	5%	DB	-10.1	-6.6	1.1	8.7	17.4	27.0	26.5	23.2	15.5	4.8	-1.7	-6.8		
		MCWB	-10.9	-7.7	-1.5	3.4	9.7	16.9	17.7	15.7	10.4	2.2	-2.9	-7.5		
	10%	DB	-13.0	-9.5	-1.3	6.1	14.4	24.9	24.5	20.8	13.1	2.7	-3.7	-9.6		
		MCWB	-13.6	-10.4	-3.4	1.9	7.9	15.9	16.7	14.7	9.1	0.9	-4.8	-10.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-4.9	-3.1	3.1	7.2	13.8	19.9	20.5	19.3	15.6	5.8	3.2	-2.7		
		MCDB	-4.1	-1.6	6.3	13.0	22.7	27.4	26.5	24.4	21.6	9.2	4.4	-1.6		
	2%	WB	-8.3	-5.0	0.7	5.4	11.5	18.7	19.3	17.8	13.2	4.1	-0.3	-5.2		
		MCDB	-7.4	-3.7	3.7	11.0	19.2	26.9	25.5	23.4	17.6	7.0	0.7	-4.3		
	5%	WB	-10.8	-7.7	-1.3	3.8	10.0	17.6	18.4	16.8	11.2	2.2	-2.9	-7.5		
		MCDB	-10.1	-6.5	1.0	8.0	16.5	25.3	24.8	21.7	14.5	4.2	-1.9	-6.8		
	10%	WB	-13.6	-10.4	-3.2	2.3	8.5	16.5	17.4	15.5	9.4	1.0	-4.8	-10.3		
		MCDB	-12.9	-9.5	-1.3	5.9	13.8	23.7	23.3	19.8	12.6	2.7	-3.7	-9.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.7	15.7	17.5	13.8	12.7	15.1	13.8	13.7	12.0	9.5	10.7	10.4		
		MCDBR	13.0	14.4	16.9	16.3	18.6	20.6	18.7	18.9	17.3	12.1	8.4	10.4		
	5% WB	MCWBR	12.4	13.0	13.7	10.8	10.5	10.1	9.3	10.6	10.8	8.8	8.0	10.0		
		MCWBR	13.1	13.8	16.2	14.6	17.4	17.8	15.8	15.5	15.8	11.0	8.3	10.6		
(40) Clear-Sky Solar Irradiance	taub		0.256	0.292	0.314	0.346	0.376	0.409	0.479	0.425	0.339	0.308	0.258	0.233		
	taud		1.976	2.011	2.053	2.115	2.283	2.292	2.071	2.196	2.438	2.298	2.028	1.925		
	Ebn at Noon		505	690	805	849	851	824	746	762	794	689	499	385		
	Edh at Noon		53	87	114	129	118	119	146	120	79	66	48	37		
(44) All-Sky Solar Radiation	RadAvg		0.55	1.52	3.06	4.61	5.18	5.73	5.23	4.12	2.54	1.22	0.64	0.32		
	RadStd		0.04	0.10	0.20	0.27	0.33	0.43	0.45	0.30	0.24	0.13	0.07	0.03		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page